

AP PHYSICS C: MECHANICS · THE UNIVERSAL STUDY GUIDE

Calculus-based — targeting a 3, 4, or 5

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

PAGE 1 | MATH OF 3/4/5 · 6 UNITS · CALCULUS DEFINITIONS

About this exam. AP Physics C: Mechanics is the calculus-based version of Physics 1. **You MUST be comfortable with derivatives + integrals** — they appear on virtually every FRQ. Topics: kinematics, Newton's laws, work/energy, momentum, rotation, gravitation, simple harmonic motion. **The exam is shorter** (90 min total) but conceptually denser.

THE MATH OF YOUR TARGET

Target	MCQ (35q, 50%)	FRQ (3q, 50%)	Approx %	Tier
3 (PASS)	~17 / 35 (~49%)	~14/45 raw	~40-44%	[3]
4	~22 / 35 (~63%)	~22/45 raw	~52-58%	[3] + [4]
5	~28 / 35 (~80%)	~32/45 raw	~70-75%	All

Time: 45 min for 35 MCQ (~1:17 each), 45 min for 3 FRQ (~15 min each). **Tight time pressure** — practice with timer.

THE 6 UNITS — AP Physics C: Mechanics (2024 redesign)

#	Unit	Weight	Focus
1	Kinematics	10–15%	1-D + 2-D motion. $\mathbf{v(t)} = \int \mathbf{a(t)} dt$, $\mathbf{x(t)} = \int \mathbf{v(t)} dt$
2	Force & Translational Dynamics	20–25%	Newton's laws, friction, drag, $\mathbf{F} = m \cdot d\mathbf{v}/dt$
3	Work, Energy, Power	15–25%	$\mathbf{W} = \int \mathbf{F} \cdot d\mathbf{r}$, U from F, conservation
4	Linear Momentum	10–20%	Impulse = $\int \mathbf{F} dt$, conservation, COM, variable mass
5	Torque & Rotational Dynamics	10–15%	$\mathbf{l} = \int \mathbf{r}^2 d\mathbf{m}$, $\tau = l\alpha$, angular momentum
6	Energy & Momentum of Rotating Systems / Oscillations / Gravitation	10–15%	Rolling, SHM, Kepler, orbital energy

THE 6 CALCULUS DEFINITIONS YOU MUST KNOW [3]/[4]

$\mathbf{v(t)} = d\mathbf{x}/dt$ — velocity is derivative of position
 $\mathbf{a(t)} = d\mathbf{v}/dt = d^2\mathbf{x}/dt^2$ — acceleration is derivative of velocity

$\mathbf{x(t)} = \int \mathbf{v(t)} dt + \mathbf{x}_0$ — position is integral of velocity
 $\mathbf{v(t)} = \int \mathbf{a(t)} dt + \mathbf{v}_0$ — velocity is integral of acceleration

$\mathbf{F(t)} = d\mathbf{p}/dt = m \cdot d\mathbf{v}/dt$ (Newton's 2nd, full form)
 Impulse: $\mathbf{J} = \int \mathbf{F} dt = \Delta\mathbf{p}$

Work: $\mathbf{W} = \int \mathbf{F} \cdot d\mathbf{r} = \int \mathbf{F(x)} dx$ (1-D)
 Power: $\mathbf{P} = d\mathbf{W}/dt = \mathbf{F} \cdot \mathbf{v}$
 Force-from-PE: $\mathbf{F} = -d\mathbf{U}/dx$ (1-D)

KEY INSIGHT [5] Identify the calculus shape on every FRQ. The exam writes problems where: (a) F or v or a is given as a function of time/position, (b) you must integrate or differentiate to get what's asked. Train yourself to ask: "Is this asking for an INTEGRAL or DERIVATIVE?" before solving.

PAGE 2 | UNIT 1: KINEMATICS — INTEGRATE & DIFFERENTIATE

Constant-Acceleration Equations [3]

$$\begin{aligned}
 v &= v_0 + at \\
 x &= x_0 + v_0 t + \frac{1}{2}at^2 \\
 v^2 &= v_0^2 + 2a(x - x_0) \\
 x - x_0 &= \frac{1}{2}(v + v_0)t
 \end{aligned}$$

Use ONLY when a is constant. For variable a , integrate.

Variable Acceleration — the Physics C signature [4]/[5]

$$\begin{aligned}
 \text{Given } a(t): \quad & v(t) = v_0 + \int_0^t a(\tau) d\tau \\
 & x(t) = x_0 + \int_0^t v(\tau) d\tau \\
 \\
 \text{Given } a(x): \quad & a = v \cdot dv/dx \quad (\text{chain rule trick} - v^2 \text{ appears}) \\
 & \frac{1}{2}(v^2 - v_0^2) = \int a(x) dx \\
 \\
 \text{Given } v(t): \quad & a(t) = dv/dt \quad \quad \quad x(t) = x_0 + \int v dt \\
 \text{Given } x(t): \quad & v(t) = dx/dt \quad \quad \quad a(t) = d^2x/dt^2
 \end{aligned}$$

2-D Motion (Projectiles) [3]/[4]

- **[3] Decompose into x and y:** x has constant velocity (no air resistance), y has constant $a = -g$.
- **[3] Initial velocity:** $v_{0x} = v_0 \cos(\theta)$, $v_{0y} = v_0 \sin(\theta)$.
- **[3] Time** is the bridge between x and y equations.
- **Range** (level ground): $R = v_0^2 \sin(2\theta) / g$. **Maximum at 45° .**

Air Resistance / Drag [4]/[5]

$$\begin{aligned}
 \text{Linear drag:} \quad & F_{\text{drag}} = -bv \quad (\text{small objects, low speed}) \\
 \text{Equation of motion (falling):} \quad & m dv/dt = mg - bv
 \end{aligned}$$

$$\text{TERMINAL VELOCITY: set } dv/dt = 0 \rightarrow v_{\text{term}} = mg / b$$

$$\text{Solution: } v(t) = v_{\text{term}} \cdot (1 - e^{(-bt/m)})$$

$$\begin{aligned}
 \text{Quadratic drag:} \quad & F_{\text{drag}} = -cv^2 \quad (\text{large objects, higher speed}) \\
 & v_{\text{term}} = \sqrt{mg/c}
 \end{aligned}$$

KEY INSIGHT [5] Drag FRQ trap. When asked "sketch $v(t)$ " for falling-with-drag: starts at $v=0$, asymptotically approaches v_{term} . NOT linear. Concavity is concave-down throughout.

PAGE 3 | UNIT 2: FORCES & TRANSLATIONAL DYNAMICS

Newton's Laws [3]

- N1: An object at rest (or constant v) stays so unless force acts.
 N2: $\Sigma F = m \cdot a$ (vector! Apply to each axis separately)
 Full form: $\Sigma F = dp/dt = d(mv)/dt$
 N3: Action-reaction: equal magnitude, opposite direction, on different objects.

Free-Body Diagrams (FBDs) — non-negotiable [3]/[4]

- [3] Draw the object as a dot. Show ALL forces with arrows from the dot.
- [3] Label each force: F_g (gravity), F_N (normal), f or F_f (friction), F_T (tension), F_{app} (applied).
- [3] Choose axes so a is along ONE axis (often tilt for incline problems).
- [4] FRQ rubric awards FBD points specifically. Even if you can't solve the problem, draw a complete FBD for partial credit.

Friction [3]/[4]

STATIC friction: $f_s \leq \mu_s \cdot N$ (just enough to prevent motion)
 KINETIC friction: $f_k = \mu_k \cdot N$ (constant magnitude when sliding)

Generally $\mu_s > \mu_k$ (harder to start than to keep going)
 Friction force OPPOSES relative motion (or attempted motion).

Inclined Planes [3]/[4]

On incline angle θ :
 Component of gravity ALONG slope (down): $mg \sin \theta$
 Component PERPENDICULAR (into surface): $mg \cos \theta$
 Normal force: $N = mg \cos \theta$
 Friction (if sliding down): $f = \mu_k \cdot mg \cos \theta$

Net acceleration down (with friction): $a = g(\sin \theta - \mu_k \cos \theta)$

Tension, Pulleys, Atwood [4]

- [4] Massless string + frictionless pulley = same tension throughout.
- [4] Atwood machine ($m_1 > m_2$): $a = (m_1 - m_2)g / (m_1 + m_2)$; $T = 2m_1m_2g / (m_1 + m_2)$.
- [4] Always pick a sign convention for each mass and stick with it.

Drag — exponential decay solution [5]

Object dropped, linear drag $F = -bv$:
 $m \, dv/dt = mg - bv$
 Separable ODE solution:
 $v(t) = (mg/b)(1 - e^{(-bt/m)}) \rightarrow v_{term} = mg/b$
 Object launched horizontally, linear drag (no gravity for x):
 $v_x(t) = v_0 \cdot e^{(-bt/m)}$

WATCH OUT [5] Recognize the differential-equation form. $dv/dt = A - Bv \rightarrow$ exponential approach to A/B . $dv/dt = -Bv \rightarrow$ exponential decay to 0.

PAGE 4 | UNIT 3: WORK · ENERGY · POWER

Work — the integral form [3]/[4]

$$\text{Work: } W = \int \vec{F} \cdot d\vec{r}$$

$$\text{CONSTANT FORCE: } W = F \cdot d \cdot \cos \theta \quad (\theta \text{ between } F \text{ and displacement})$$

$$\text{VARIABLE FORCE: } W = \int F(x) dx \quad (\text{area under } F\text{-vs-}x \text{ graph})$$

SIGN CONVENTIONS:

F and motion same direction: $W > 0$ (force does POSITIVE work)F perpendicular to motion: $W = 0$ (e.g., normal force, circular motion)F opposes motion: $W < 0$ (e.g., friction, decelerating force)

Kinetic Energy + Work-Energy Theorem [3]/[4]

$$KE = \frac{1}{2}mv^2$$

$$\text{Work-Energy Theorem: } W_{\text{NET}} = \Delta KE = \frac{1}{2}m(v^2 - v_0^2)$$

$$\text{For SPRING: } W_{\text{spring}} = \frac{1}{2}k(x_i^2 - x_f^2) \quad (\text{REVERSED — work done BY spring})$$

(negative of ΔU_{spring})

Potential Energy & Conservative Forces [3]/[4]/[5]

CONSERVATIVE FORCE has a potential energy U:

$$F = -dU/dx \quad (\text{in 1-D})$$

$$\vec{F} = -\nabla U \quad (\text{in 3-D})$$

$$\Delta U = -W_{\text{conservative}}$$

$$\text{GRAVITATIONAL (near surface): } U_g = mgh$$

$$\text{GRAVITATIONAL (general): } U_g = -GMm/r$$

$$\text{SPRING: } U_s = \frac{1}{2}kx^2$$

$$\text{Force from } U \text{ (1-D): } F(x) = -dU/dx$$

$$\text{Equilibrium points: } dU/dx = 0$$

$$\text{Stable: } d^2U/dx^2 > 0 \quad (U \text{ is min}) \quad \text{Unstable: } d^2U/dx^2 < 0 \quad (U \text{ is max})$$

Conservation of Mechanical Energy [3]/[4]

$$\text{No non-conservative forces: } KE_i + U_i = KE_f + U_f$$

With friction or other dissipation:

$$KE_i + U_i = KE_f + U_f + |W_{\text{friction}}|$$

$$\text{or: } \Delta KE + \Delta U = W_{\text{nonconservative}}$$

Power [3]/[4]

$$\text{Average power: } P_{\text{avg}} = W / \Delta t$$

$$\text{Instantaneous: } P = dW/dt = \vec{F} \cdot \vec{v} = F \cdot v \cdot \cos \theta$$

$$\text{Units: watt (W) = J/s = kg} \cdot \text{m}^2/\text{s}^3$$

KEY INSIGHT [5] U-vs-x graph reading. Steep negative slope $\rightarrow$ large positive force in +x. Local minimum $\rightarrow$ stable equilibrium $\rightarrow$ SHM near min. Local maximum $\rightarrow$ unstable equilibrium. Total energy line crossing U $\rightarrow$ turning points. This is a frequent FRQ.

PAGE 5 | UNIT 4: MOMENTUM · IMPULSE · CENTER OF MASS

Momentum & Impulse [3]/[4]

$$\vec{p} = m \cdot \vec{v} \quad (\text{vector})$$

$$\text{Newton's 2nd (full): } \vec{F} = d\vec{p}/dt$$

$$\text{IMPULSE: } \vec{J} = \int \vec{F} dt = \Delta \vec{p}$$

On F-vs-t graph, impulse = AREA

$$\text{Avg force: } F_{\text{avg}} = J/\Delta t$$

Conservation of Momentum — when applicable [3]/[4]

- [3] When net EXTERNAL force = 0 during interaction (collision, explosion), total momentum is conserved.
- [3] Conserved separately in x and y. 2-D problems: write p_x and p_y equations independently.
- [4] During collisions, internal forces are large but cancel by N3 — only externals matter, and they're usually negligible during the brief collision time.

Collision Types [3]/[4]

Type	Conserves	Identifying clue
Elastic [3]	p AND KE conserved	Bouncing balls, gas molecules, billiards (idealized)
Inelastic [3]	p only (KE lost to heat/sound/deformation)	Most real collisions
Perfectly inelastic [3]/[4]	p only; objects STICK together	Bullet embeds in block, train cars couple

$$\text{PERFECTLY INELASTIC (1-D): } m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$$

ELASTIC (1-D): Use BOTH p and KE conservation. Two unknowns, two equations.

Special cases: $m_1 = m_2 \rightarrow$ exchange velocities

Equal mass + one at rest $\rightarrow$ moving mass STOPS, other gets v

Center of Mass [4]/[5]

$$\text{DISCRETE: } x_{\text{CM}} = (\sum m_i x_i) / (\sum m_i)$$

$$\text{CONTINUOUS: } x_{\text{CM}} = (\int x dm) / M$$

$$\text{COM motion: } v_{\text{CM}} = (\sum m_i v_i) / M$$

The CM moves as if all mass were concentrated there: $F_{\text{ext}} = M \cdot a_{\text{CM}}$

EXAMPLE — uniform rod length L from $x=0$ to $x=L$:

$$dm = (M/L) dx \quad \rightarrow \quad x_{\text{CM}} = L/2$$

Variable Mass — rocket [5]

Rocket equation (no gravity):

$$m dv/dt = -u (dm/dt) \quad u = \text{exhaust velocity (relative)}$$

$$\Delta v = u \cdot \ln(m_0/m_f) \quad (\text{Tsiolkovsky equation})$$

PAGE 6 | UNIT 5: ROTATION — TORQUE · MOMENT OF INERTIA · ANGULAR MOMENTUM**Rotational Kinematics — perfect analogy to linear [3]/[4]**

LINEAR ↔ ANGULAR
 $x \leftrightarrow \theta$ (radians!)
 $v \leftrightarrow \omega = d\theta/dt$
 $a \leftrightarrow \alpha = d\omega/dt$

Constant α equations (analog):

$$\begin{aligned}\omega &= \omega_0 + \alpha t \\ \theta &= \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 \\ \omega^2 &= \omega_0^2 + 2\alpha(\theta - \theta_0)\end{aligned}$$

TANGENTIAL relations:

$$v_t = r\omega \quad a_t = r\alpha \quad a_{\text{centripetal}} = v^2/r = r\omega^2$$

Moment of Inertia [3]/[4]/[5]

DISCRETE: $I = \sum m_i r_i^2$
 CONTINUOUS: $I = \int r^2 dm$ (signature C-Mech FRQ!)

COMMON I'S (axis through center of mass):

$$\begin{aligned}\text{Solid disk / cylinder:} & I = \frac{1}{2}MR^2 \\ \text{Solid sphere:} & I = \frac{2}{5}MR^2 \\ \text{Hollow sphere (shell):} & I = \frac{2}{3}MR^2 \\ \text{Thin ring / hoop:} & I = MR^2 \\ \text{Rod about CM (length L):} & I = \frac{1}{12}ML^2 \\ \text{Rod about end:} & I = \frac{1}{3}ML^2\end{aligned}$$

PARALLEL AXIS THEOREM: $I = I_{\text{CM}} + Md^2$
 (d = distance from CM to new parallel axis)

KEY INSIGHT [5] FRQ favorite: derive I from integration. Set up $dm = (M/L) dx$ (rod) or $dm = (M/\pi R^2)(2\pi r) dr$ (disk via rings). Compute $I = \int r^2 dm$ with proper limits. Common derivation problems: rod, disk, cone, sphere.

Torque & Newton's 2nd for Rotation [3]/[4]

$$\tau = r \times F \quad (\text{vector cross product}) \quad |\tau| = rF \sin \theta$$

$$\begin{aligned}\Sigma \tau &= I \cdot \alpha \quad (\text{rotational analog of } F = ma) \\ \Sigma \tau &= dL/dt \quad (\text{general form})\end{aligned}$$

$$\begin{aligned}\text{WORK and POWER:} & W = \int \tau d\theta \quad P = \tau \cdot \omega \\ \text{Rotational KE:} & KE_{\text{rot}} = \frac{1}{2} I \omega^2\end{aligned}$$

Angular Momentum & Conservation [4]/[5]

$$\begin{aligned}L &= I\omega \quad (\text{rigid body about fixed axis}) \\ \vec{L} &= \vec{r} \times \vec{p} \quad (\text{point mass})\end{aligned}$$

$$\Sigma \tau_{\text{ext}} = dL/dt$$

$$\text{CONSERVATION: if } \Sigma \tau_{\text{ext}} = 0 \rightarrow L_i = L_f$$

CLASSIC: figure skater pulls in arms $\rightarrow I$ decreases $\rightarrow \omega$ increases.
 $I\omega$ stays constant (until external τ acts).

Rolling Without Slipping [4]/[5]

$$\text{CONDITION: } v_{\text{CM}} = r\omega, \quad a_{\text{CM}} = r\alpha$$

$$\text{KE of rolling object: } KE = \frac{1}{2}M v_{\text{CM}}^2 + \frac{1}{2}I_{\text{CM}} \omega^2 = \frac{1}{2}(M + I/r^2) v^2$$

Rolling DOWN incline of angle θ (no slip):

$$a = g \sin \theta / (1 + I/(MR^2))$$

Rank by speed at bottom (race): hoop (slowest) < disk < solid sphere < block (no rotation, fastest)

PAGE 7 | UNIT 6: SHM · GRAVITATION

Simple Harmonic Motion [3]/[4]

Defining condition: $F = -kx \rightarrow a = -(k/m)x$
 Differential eq: $d^2x/dt^2 = -\omega^2 x$ where $\omega = \sqrt{k/m}$

Solution: $x(t) = A \cos(\omega t + \phi)$
 $v(t) = -A\omega \sin(\omega t + \phi)$ $\max |v| = A\omega$ (at $x = 0$)
 $a(t) = -A\omega^2 \cos(\omega t + \phi)$ $\max |a| = A\omega^2$ (at $x = \pm A$)

Period: $T = 2\pi/\omega = 2\pi\sqrt{m/k}$

Frequency: $f = 1/T = \omega/(2\pi)$

Energy: $E = \frac{1}{2}kA^2$ (constant) = $\frac{1}{2}kx^2 + \frac{1}{2}mv^2$ (always)

Pendulum (small angle) [3]/[4]

SIMPLE PENDULUM (point mass at end of length L):
 $T = 2\pi \sqrt{L/g}$ (independent of mass!)

PHYSICAL PENDULUM (any rigid body, pivot at distance d from CM):
 $T = 2\pi \sqrt{I / (Mgd)}$

Both valid only for small θ ($\sin \theta \approx \theta$).

Gravitation — Universal Law [3]/[4]

Universal: $F = G m_1 m_2 / r^2$ $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$

Gravitational PE (general, with $U \rightarrow 0$ as $r \rightarrow \infty$):
 $U = -G M m / r$

Surface g (planet of mass M , radius R):
 $g = GM/R^2$

Orbits — Circular Motion in Gravity [4]/[5]

CIRCULAR ORBIT: gravity provides centripetal force
 $GMm/r^2 = mv^2/r \rightarrow v_{\text{orbit}} = \sqrt{GM/r}$

Period (Kepler's 3rd): $T^2 = 4\pi^2 r^3 / (GM)$ $\rightarrow T^2 \propto r^3$

TOTAL ENERGY of circular orbit:
 $E = KE + U = \frac{1}{2}mv^2 + (-GMm/r) = -GMm/(2r)$ (NEGATIVE: bound)

ESCAPE VELOCITY ($E = 0$ at infinity):
 $v_{\text{esc}} = \sqrt{2GM/R}$ (= $\sqrt{2} \times v_{\text{orbit}}$ at surface)

KEY INSIGHT [5] Energy hierarchy of orbits. $E < 0$ means BOUND (circle or ellipse). $E = 0$ means PARABOLIC (just escapes). $E > 0$ means HYPERBOLIC (escapes with KE remaining). For a circular orbit specifically, $KE = \frac{1}{2}|U|$ and $E = -KE = U/2$.

Kepler's Laws [3]/[4]

- **1st [3]:** planets move in ELLIPSES with sun at one focus.
- **2nd [3]/[4]:** line from sun sweeps EQUAL AREAS in equal times $\rightarrow$ conservation of angular momentum.
- **3rd [3]/[4]:** $T^2 \propto a^3$ (a = semi-major axis).

PAGE 8 | MASTER INTEGRATION TABLE — every C-Mech FRQ pattern

This page is your secret weapon. AP Physics C: Mech FRQs are built from a small set of integration / differentiation patterns. Recognize the pattern → set up the integral → done.

PATTERN 1: Find $v(t)$ or $x(t)$ from $a(t)$

Given $a(t)$, x_0 , v_0 :

$$v(t) = v_0 + \int_0^t a(\tau) d\tau$$

$$x(t) = x_0 + \int_0^t v(\tau) d\tau$$

EXAMPLE: $a(t) = 6t$. $v_0 = 0$, $x_0 = 0$.

$$v(t) = 0 + \int_0^t 6\tau d\tau = 3t^2$$

$$x(t) = 0 + \int_0^t 3\tau^2 d\tau = t^3$$

PATTERN 2: Work from variable force $F(x)$

$$W = \int_{x_a}^{x_b} F(x) dx$$

EXAMPLE: $F(x) = ax + bx^2$. Work from $x=0$ to $x=L$?

$$W = \int_0^L (ax + bx^2) dx = \frac{1}{2}aL^2 + \frac{1}{3}bL^3$$

PATTERN 3: Impulse from $F(t)$

$$J = \int_0^t F(\tau) d\tau = \Delta p = m \cdot \Delta v$$

EXAMPLE: $F(t) = F_0$ for $0 < t < T$, else 0. Mass m starts at rest.

$$J = F_0 \cdot T \rightarrow \Delta v = F_0 T / m$$

PATTERN 4: Center of Mass via integration

$$x_{CM} = (1/M) \int x dm \quad \text{where } dm = \lambda(x) dx$$

EXAMPLE: Rod with linear density $\lambda(x) = ax$, length L .

$$M = \int_0^L ax dx = \alpha L^2 / 2$$

$$x_{CM} = (1/M) \int_0^L x \cdot ax dx = (\alpha L^3 / 3) / (\alpha L^2 / 2) = 2L/3$$

PATTERN 5: Moment of Inertia $I = \int r^2 dm$

ROD about end (length L , mass M , uniform):

$$dm = (M/L) dx$$

$$I = \int_0^L x^2 (M/L) dx = ML^2/3$$

DISK about center (radius R , mass M , uniform, thickness negligible):

$$dm = (M/\pi R^2) \cdot 2\pi r dr = (2M/R^2) r dr$$

$$I = \int_0^R r^2 \cdot (2M/R^2) r dr = \frac{1}{2}MR^2$$

Always: pick a coordinate, write dm in terms of it, set up $r^2 dm$, integrate.

PATTERN 6: Solve $dv/dt = A - Bv$ (drag)

Separable: $dv / (A - Bv) = dt$

Integrate: $-(1/B) \ln|A - Bv| = t + C$

Solve: $v(t) = (A/B) - [(A/B) - v_0] \cdot e^{(-Bt)}$

Asymptote (terminal velocity): $v \rightarrow A/B$ as $t \rightarrow \infty$

PATTERN 7: $F = -dU/dx \rightarrow$ equilibrium analysis

Given $U(x)$:

Equilibrium points: $dU/dx = 0$

Stable equilibrium: $d^2U/dx^2 > 0$ (U is local min)

Unstable: $d^2U/dx^2 < 0$ (U is local max)

Near a stable point: $U \approx \frac{1}{2}k(x - x_0)^2$ where $k = d^2U/dx^2$ at x_0

→ SHM with $\omega = \sqrt{k/m}$

PAGE 9 | FRQ STRATEGY · COMMON TRAPS · CALCULATOR USE

The 3 FRQs — predictable structure [3]/[4]

Each FRQ ~15 points, often with 5-7 subparts (a, b, c, ...). At least one FRQ is a **lab/experimental design** question. At least one involves **deriving an expression** symbolically.

FRQ type	Recipe
Symbolic derivation	Set up $F = ma$ (or $\tau = I\alpha$, or energy conservation), substitute given quantities, solve for the requested variable IN SYMBOLS. Don't plug numbers until asked.
Calculus integration	Recognize the pattern (above). Set up integral with proper limits. Evaluate. Show every step.
Lab / experimental	Identify what to measure, how to plot data to get a LINEAR graph (often log-log or $1/T^2$ vs L), how to extract requested quantity from slope/intercept.

FRQ rubric earners — what graders look for [3]/[4]/[5]

- [3] **Free-body diagram** if forces are involved. Often worth 1-3 points alone.
- [3] **Define your sign convention.** "Take down as positive."
- [3] **Show fundamental equation BEFORE manipulating.** "From Newton's 2nd: $\Sigma F = ma \rightarrow mg \sin \theta - \mu mg \cos \theta = ma$ ".
- [4] **Carry symbols, plug numbers last.** Earns derivation points + reduces arithmetic mistakes.
- [4] **Units on every numerical answer.** Missing units = lose 1 point per answer typically.
- [5] **Sketch graphs cleanly:** label axes with VARIABLE + units, mark important values (intercept, asymptote, critical points), show scale.
- [5] **Sanity-check answers.** Limit cases: "if $M = 0$, this should reduce to...". Catches sign errors + algebra mistakes.

Top calculus / physics traps [3]/[4]/[5]

- [3] **Confusing $v^2 = v_o^2 + 2a\Delta x$ with constant-a only.** Don't use kinematic equations when a is variable — integrate.
- [3] **Using mgh for U at large distance.** mgh works only NEAR Earth's surface. For orbits, use $U = -GMm/r$.
- [4] **Forgetting r^2 (not r) in moment of inertia integral.**
- [4] **Spring PE sign:** $U_{\text{spring}} = \frac{1}{2}kx^2$ (ALWAYS positive when stretched OR compressed).
- [5] **Rolling without slipping:** must include BOTH translational AND rotational KE. Many students forget the rotational piece.
- [5] **Gravitational orbit energy sign:** bound orbits have NEGATIVE total energy. $KE = -E = +GMm/(2r)$ for circular.
- [5] **Drag exponential:** v_{term} is APPROACHED, not reached. $v(t)$ is asymptotic, not linear or quadratic.

Calculator use [3]/[4]

- [3] **Calculator allowed on both MCQ and FRQ.** TI-84 graphing calculator is standard.
- [4] **Use NUMERICAL integration ($\int f(x)dx$) on calculator** for hard integrals on MCQ.
- [4] **Solve(...) for transcendental equations** (e.g., v_{term} equations). Faster than algebra.
- [5] **DON'T calculator-bash on FRQ.** Show analytical work — graders need to see the physics, not just an answer.

PAGE 10 | SCORE LADDER · TIER CHECKLISTS · THE ONE THING

THE SCORE LADDER

For a 3 (Pass)	For a 4	For a 5
MCQ: 17-21 / 35	MCQ: 22-26 / 35	MCQ: 28-32 / 35
Constant-a kinematics, $F=ma$, mgh , $\frac{1}{2}mv^2$, $\frac{1}{2}kx^2$ — basics solid.	Add: variable a (integrate), work integrals, drag, parallel axis.	Full mastery: COM/I derivations, drag ODE, orbital energy, SHM derivations.
Draw FBD on every force problem.	Energy conservation with friction, perfectly inelastic collisions.	Rolling, angular momentum conservation (skater), Kepler's 3rd.
Recognize when to use kinematics vs energy.	Recognize when to integrate vs use constant-a.	ODE setups for variable-mass + drag problems; sketch $v(t)$, $U(x)$.

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — focus only on these:

- ✓ MCQ: 17+ correct.
- ✓ Free-body diagram on every force FRQ part.
- ✓ Constant-acceleration kinematics solid (4 equations).
- ✓ $KE + U = \text{const}$ for energy problems without friction.
- ✓ Conservation of momentum in collisions.

For a 4 — add these on top:

- ✓ MCQ: 22+ correct.
- ✓ Variable acceleration: integrate $a(t) \rightarrow v(t) \rightarrow x(t)$.
- ✓ Variable force: $W = \int F(x) dx$.
- ✓ Linear drag: terminal velocity, basic ODE solution.
- ✓ Rotation: $\tau = I\alpha$, basic moments of inertia, $\frac{1}{2}I\omega^2$.
- ✓ SHM: $T = 2\pi\sqrt{m/k}$; pendulum $T = 2\pi\sqrt{L/g}$.

For a 5 — add these on top:

- ✓ MCQ: 28+ correct.
- ✓ $I = \int r^2 dm$ derivation for any common shape.
- ✓ $U(x)$ analysis: equilibrium points + stability + small-oscillation period.
- ✓ Rolling without slipping: BOTH translational + rotational KE.
- ✓ Orbital energy: $E = -GMm/(2r)$, escape velocity = $\sqrt{2} \times \text{orbit velocity}$.
- ✓ Drag ODE solution: $v(t) = v_{\text{term}}(1 - e^{-(bt/m)})$.
- ✓ Symbolic derivations carrying letters until last step; check limit cases.

THE ONE THING THAT MATTERS MOST [3][4][5]: ALWAYS ASK "INTEGRAL OR DERIVATIVE?" BEFORE PUSHING ALGEBRA. AP Physics C: Mech rewards students who recognize the calculus pattern instantly. Variable $a \rightarrow$ integrate. Variable $F \rightarrow$ integrate. Find v from $x \rightarrow$ differentiate. Find equilibrium from $U \rightarrow$ set derivative to zero. **Make this your reflex.** Once the calculus is set up correctly, the physics-arithmetic is trivial. Most students lose points by trying to use constant-acceleration kinematics on variable-acceleration problems.

If you do nothing else: master the 7 integration patterns on Page 8, draw FBDs religiously, and carry symbols until the final step.

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